

Education

2022–2027 (expected)	University of Southern California <i>Ph.D. in Computational Biology and Biomedical Informatics</i> Dr. Michael ‘Doc’ Edge, advisor	Los Angeles, CA
2016–2020	Amherst College <i>B.A. in Computer Science</i> <i>B.A. in Sexuality, Women’s and Gender Studies</i> Cumulative GPA: 3.8	Amherst, MA
2019	Yale-National University of Singapore <i>International Exchange Program</i>	Singapore

Research and Professional Experience

2022– Present	Graduate Research Assistant <i>Dr. Michael ‘Doc’ Edge, USC Department of Quantitative Computational Biology</i> Developing and applying population-genetic methods to improve forensic identity inference.	Los Angeles, CA
2020–2022	Postgraduate Research Associate <i>Dr. Jeffrey Ishizuka, Yale University School of Medicine</i> Analyzed multiplexed tumor data to reveal immunotherapy resistance-associated mutations.	New Haven, CT
2019–2020	Research Assistant <i>Dr. Kate Follette and Dr. Matteo Riondato, Amherst College</i> Developed a database to support direct imaging and characterization of exoplanets.	Amherst, MA
2017-2020	Teaching Assistant and Ginsburg Fellow <i>Amherst College Computer Science Department</i> Collaborated with faculty to lead computational labs and integrate algorithmic thinking into mathematics instruction.	Amherst, MA

Publications

5. **Ouerghi, F.**, Krane, D. E., & Edge, M. D. (in press). *On forensic likelihood ratios from low-coverage sequencing*. Forensic Science International: Genetics.
4. Perry, C., Frey, A., Ton, W., Mackie, M., **Ouerghi, F.**, Wei, J., Clulo, K., He, M., Dumit, T. C., Ding, M., Guo, W., Clune, J., Olino, K., & Ishizuka, J. (2023). 1107 Enhancing immune responses to melanoma with the RIG-I antiviral pathway agonist SLR14. *Journal for ImmunoTherapy of Cancer*, 11(Suppl 1).
3. Horowitch, B., Lee, D. Y., Ding, M., Martinez-Morilla, S., Aung, T. N., **Ouerghi, F.**, Wang, X., Wei, W., Damsky, W., Sznol, M., Kluger, H., Rimm, D. L., & Ishizuka, J. J. (2023). Subsets of IFN Signaling Predict Response to Immune Checkpoint Blockade in

Patients with Melanoma. *Clinical Cancer Research*, 29(15), 2908–2918.

2. Lee, D., Horowitch, B., Ding, M., Martinez-Morilla, S., Aung, T. N., **Ouerghi, F.**, Wang, X., Wei, W., Damsky, W., Szol, M., Kluger, H. M., Rimm, D. L., & Ishizuka, J. J. (2023). Subsets of interferon signaling and response to immune checkpoint blockade. *Journal of Clinical Oncology*, 41(16_suppl), 9522–9522.

1. Dong, M., Wang, B., Wei, J., de O. Fonseca, A. H., Perry, C. J., Frey, A., **Ouerghi, F.**, Foxman, E. F., Ishizuka, J. J., Dhodapkar, R. M., & van Dijk, D. (2023). Causal identification of single-cell experimental perturbation effects with CINEMA-OT. *Nature Methods*, 20(11), 1769–1779.

Awards and Fellowships

2025	USC Dornsife Ph.D. Academy Scholarship and Research Fund Award
2024, 2025	USC Women in STEM and Engineering Association Travel Grant
2022–2027	USC Provost’s Fellowship
2022–2025	John Woodruff Simpson Fellowship for Graduate Studies
2019	Lyle A. McGeoch Computer Science and Algorithmic Thinking Fellowship
2017, 2019	Loeb Center for Career Exploration and Planning Summer Award
2018	Dean of the Faculty Award for Grace Hopper Conference
2018	Center for Community Engagement Summer Award

Conference Presentations

June 2025	Modeling and Theory in Population Biology “On forensic likelihood ratios from low-coverage sequencing data” (poster)	Chicago, IL
Nov 2024	USC Quantitative and Computational Biology Department Retreat “On forensic likelihood ratios from low-coverage sequencing data” (poster)	Ventura, CA
July 2024	UCLA Computational Genomics Summer Institute “How should we report genetic matches following an investigative genetic genealogy search?” (invited talk)	Los Angeles, CA
July 2024	Society for Molecular Biology and Evolution “How should we report genetic matches following an investigative genetic genealogy search?” (talk)	Puerto Vallarta, Mexico
Aug 2021	Yale Postgraduate Trainee Symposium “An integrative single-cell framework for insights into the tumor microenvironment in response to therapeutic perturbations” (poster)	New Haven, CT

Academic Service and Leadership

2024– Present	USC Graduate Student Government <i>Grant Review Committee Member</i>
2024– Present	Project SHORT Mentorship Program <i>Ph.D. Admissions Mentor</i>
2023– Present	USC Edge Lab <i>Undergraduate Research Mentor</i>
2022– Present	USC Quantitative Computational Biology Retreat <i>Co-Organizer</i>
2023– Present	USC Women in Science and Engineering Association <i>Undergraduate Mentor</i>
2023– Present	USC Computational Biology Student Association <i>First-Year Ph.D. Student Mentor</i>
2020–2022	Yale School of Medicine Ishizuka Lab <i>Undergraduate Research Mentor</i>